42115 Network Optimization

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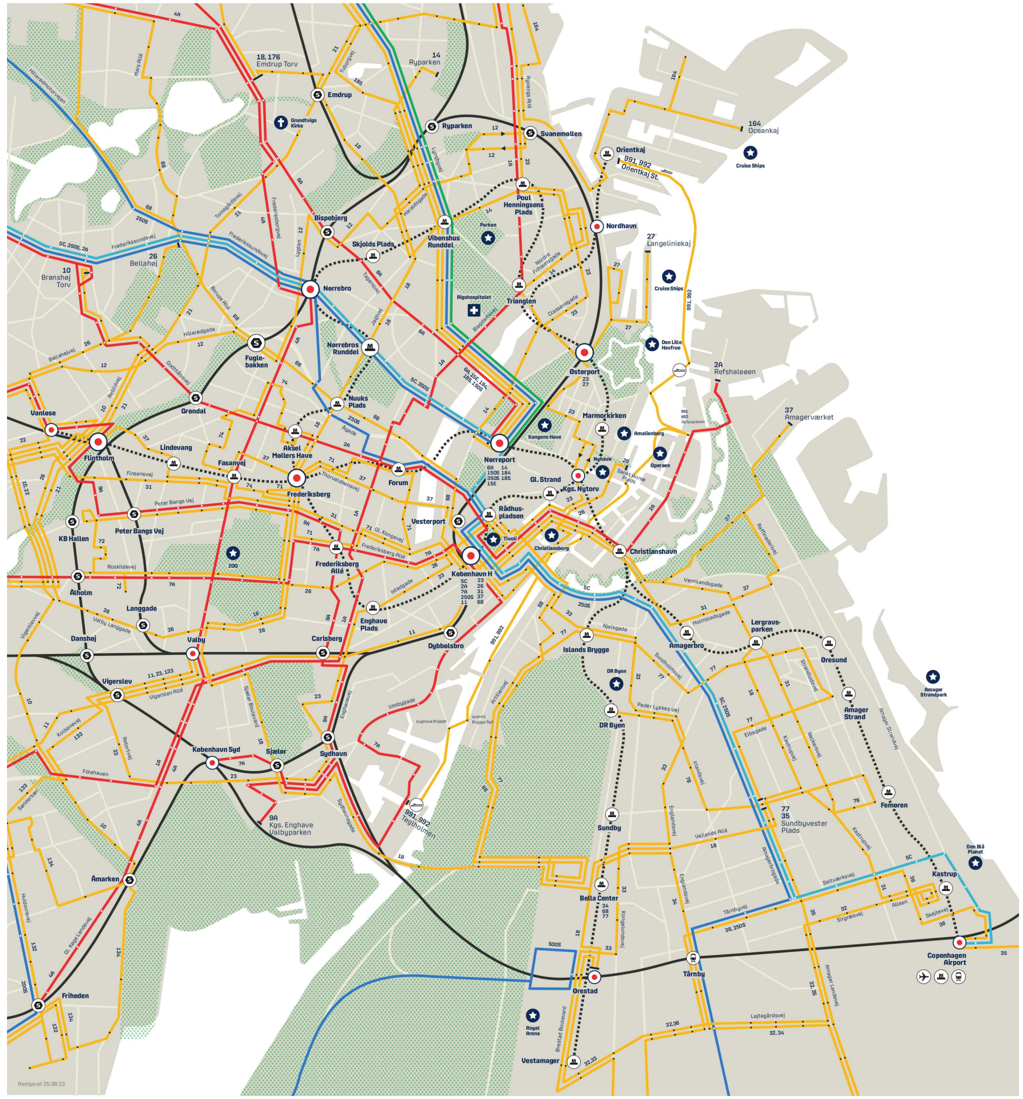
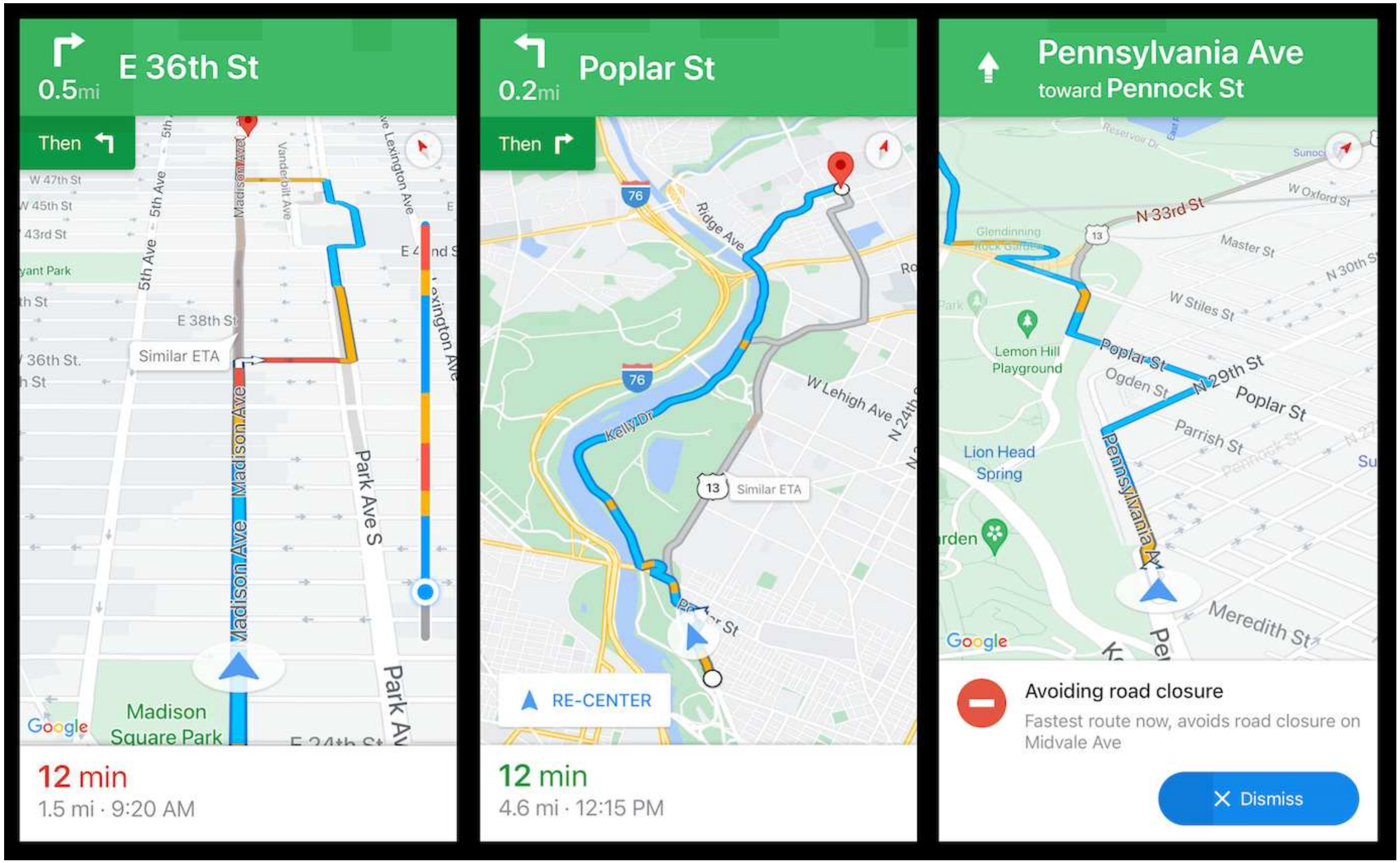
Network Examples



Bus/metro network

Network Examples



Finding way (google maps)

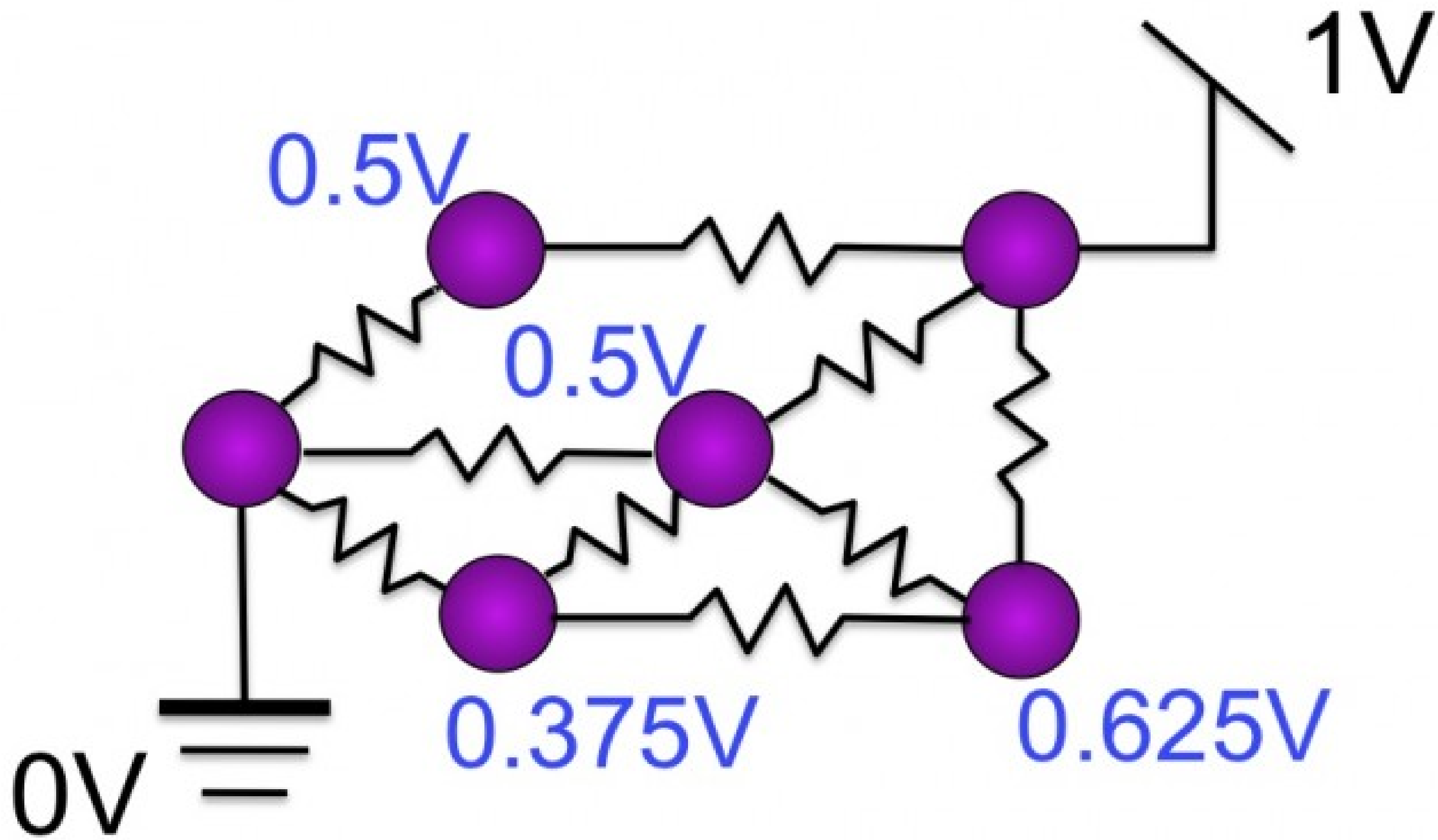
shortest path, cheapest path time constraint, alternative path

Network Examples

Telecommunication network

network design, routing data

Network Examples



Electrical network

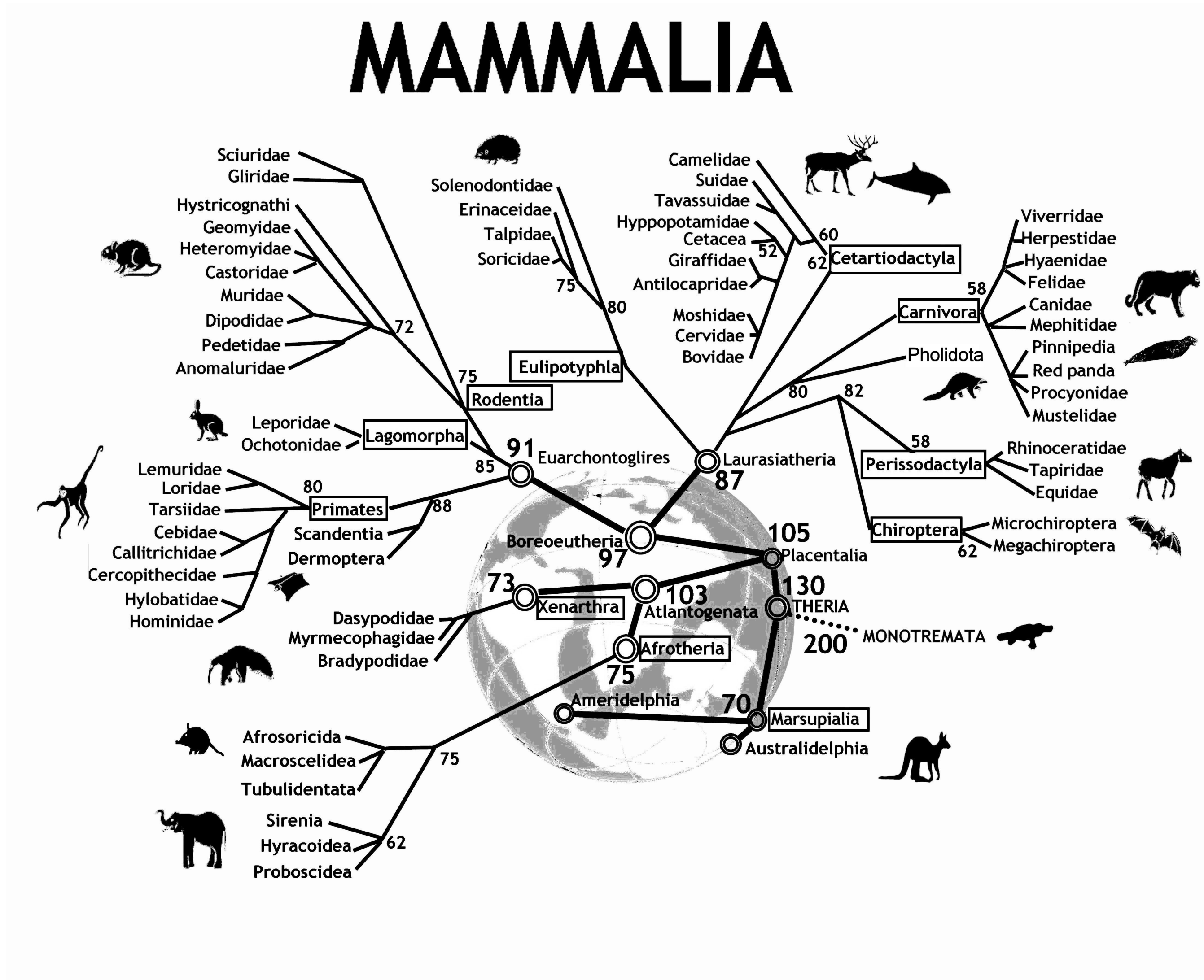
current flows (quadratic MDF)

Network Examples

Relation network (e.g. linkedin)

distance to person, connectivity

Network Examples



Evolution network

network design, distance

Motivation

- Network problems some of the most frequently occuring optimization problems
- Network models easy to model, intuitive to understand
- Many network problems can be solved efficiently
- Network problems appear as subproblem (e.g. shortest path subproblem in manpower planning)

Motivation

Complements course 42114 Integer Programming

- Use networks to model problems instead of constr. $Ax =$
- Develope efficient network algorithms where integer programming is hard to solve
- Use knowledge from LP to design algorithms
- Use knowledge from LP to prove that algorithms work correct

Course Plan

- **5 sept** Introduction, critical path, project planning
- **12 sept** Transportation problem, network simplex, linear programming
- **19 sept** Fixed-charge transportation problem, decomposition, column generation
- **26 sept** Shortest path problem, threshold partitioned shortest path algorithm
- **3 oct** Resource constrained shortest path problem
- **10 oct** Maximum flow problem I
- **17 oct — fall vacations —**
- **24 oct** Maximum flow problem II
- **31 oct** Minimum cost flow I
- **7 nov** Minimum cost flow II
- **14 nov** Multicommodity flow I
- **21 nov** Multicommodity flow II
- **28 nov** Project assignment
- **5 dec** Project assignment

Literature

- Chapters from Larsen, Clausen "Supplementary Notes"
- Chapters from Ahuja, Orlin, Magnanti "Network Flows"

Available on campusnet (DTU inside)

Exam

- Project assignment - split into two parts
- Project in groups 2-3 people

- Written exam, 17. december
- Individual written exam 4 hours, books/computer allowed, no internet

- Project must be passed to attend exam
- Passed project gives acces to 3 following exams (December, June, December)

Expectations

5 ECTS corresponds to **9 hours** of work per week

- At home: read text (briefly) before the lecture
- Attend lecture, exercises
- At home: solve last exercises and read text more carefully

You cannot expect to pass course if you only work 4 hours per week

Expectations

- Come to the lectures - video recordings are intended as a backup in case you are sick
- Come to the exercises - I do not have time to answer questions on email
- No plagiarism in project assignment - every year I report 1-2 groups

Modeling network problems

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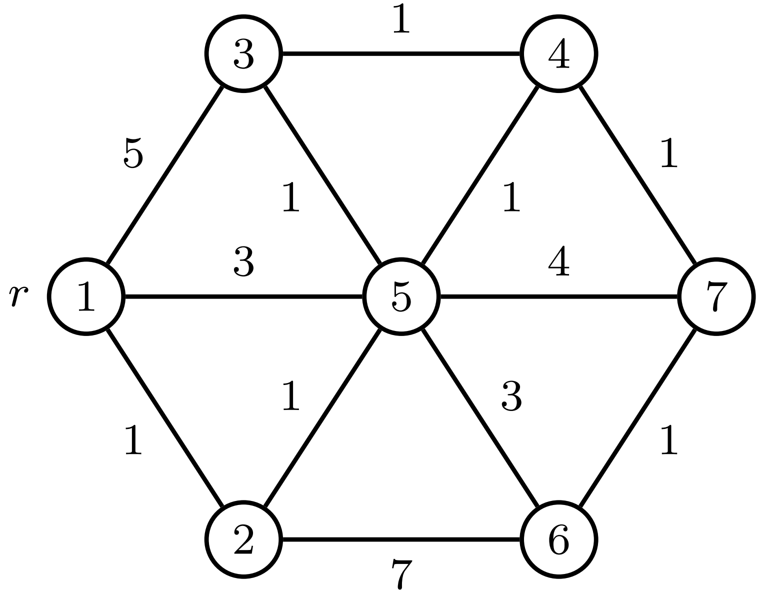
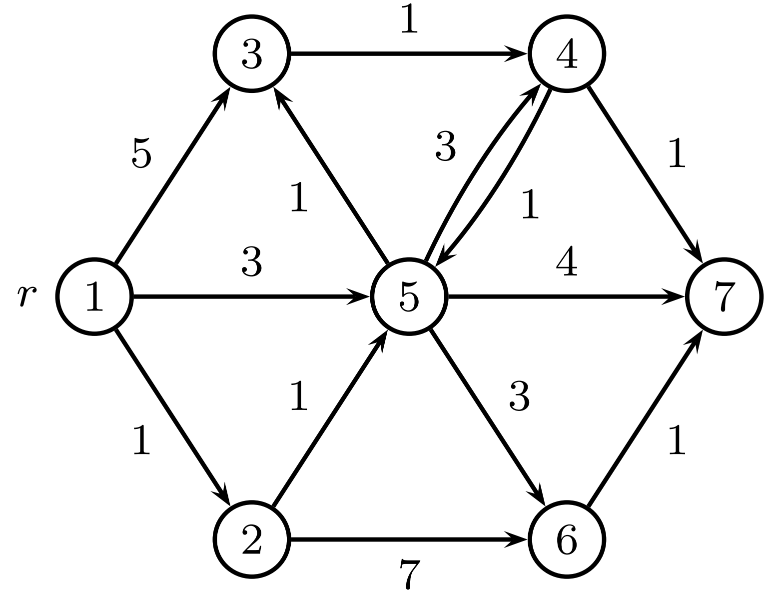
Modeling network problems

Graph (V, E)



Directed graph (V, E) *Digraph*



- Special graphs: path, cycle, tree
- Edge weights: cost, capacity, time
- Node weights: cost, time
- Special nodes: (root r, sink s), (source s, terminal t)

LP-models

Variable x_{ij} for each edge (i, j).

For path problems, x_{ij} is decision variable

$$x_{ij} = \begin{cases} 1 & \text{if edge } (i,j) \text{ is chosen} \\ 0 & \text{otherwise} \end{cases}$$

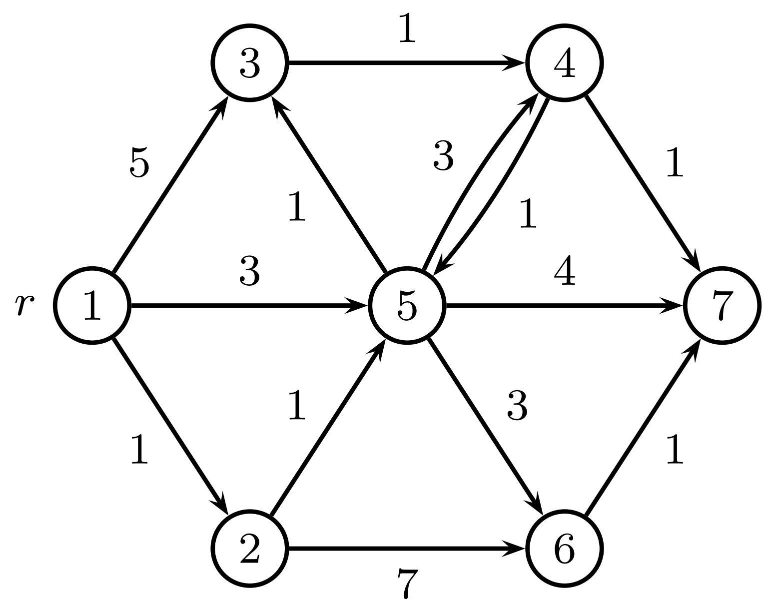
For flow problems, x_{ij} is flow on edge (i, j)

$$x_{ij} \geq 0$$

Capacity

If an edge (i, j) has a capacity c_{ij} then

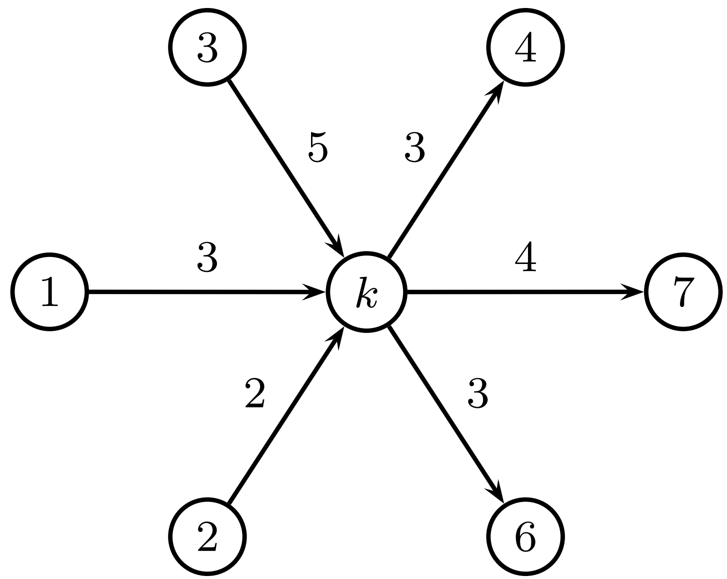
$$0 \leq x_{ij} \leq c_{ij}$$



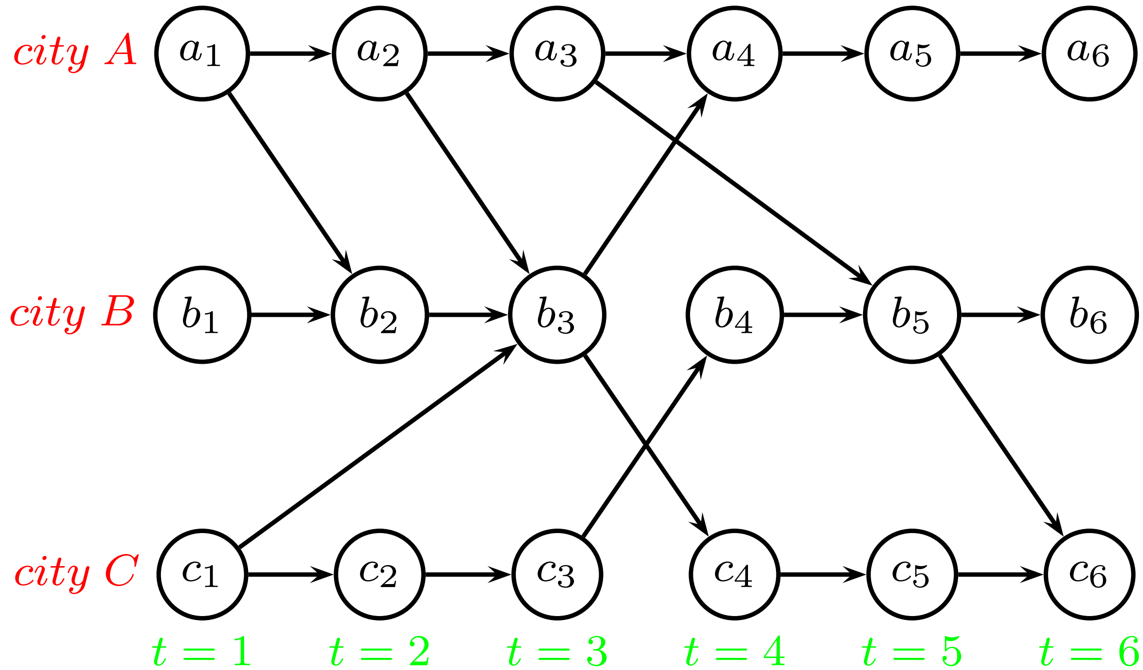
Flow conservation

In every node k: What flows in must flow out

$$\sum_{i \in V} x_{ik} = \sum_{i \in V} x_{ki}$$



Time-space graph



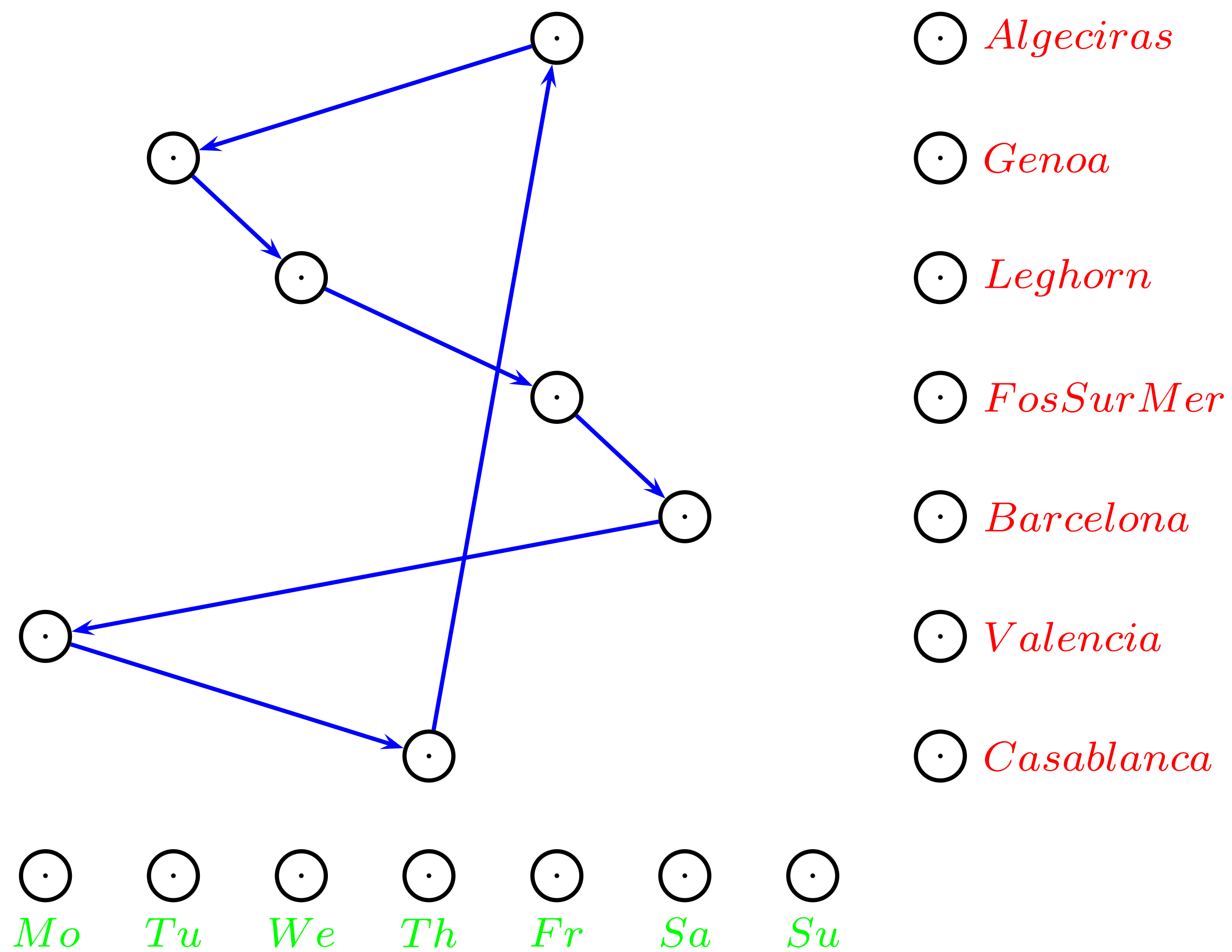
Cyclic time-space graph

LIGURIAN SEA

Port	Arrives	Departs	Transit
Algeciras, Spain	FRI	SAT	-
Algeciras, Spain	SAT	SUN	1
Genoa, Italy	TUE	WED	5
Leghorn, Italy	WED	THU	6
Fos Sur Mer, France	FRI	FRI	8
Barcelona, Spain	SAT	SUN	9
Valencia, Spain	MON	MON	10
Valencia, Spain	MON	TUE	11
Casablanca, Morocco	THU	THU	13

Cyclic time-space graph

Arrival days (first day):

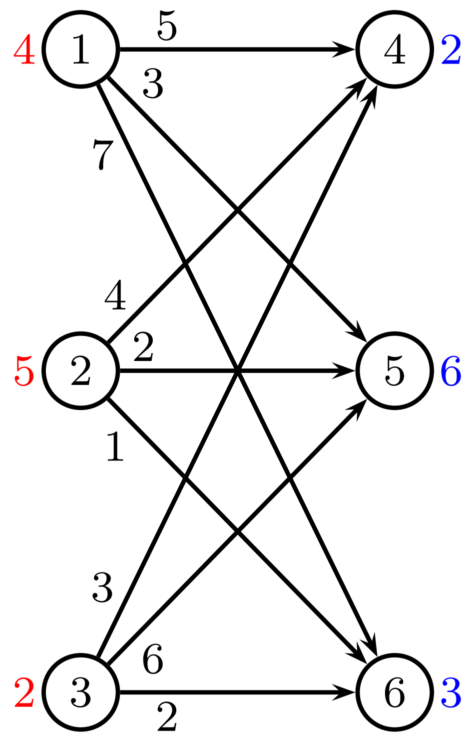


Problems

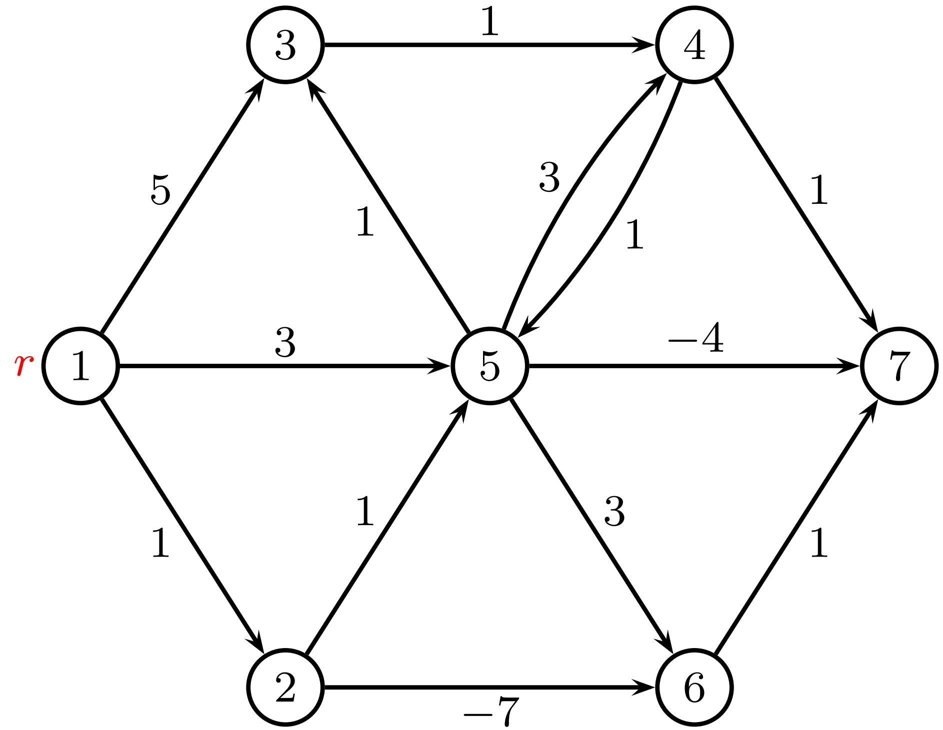
Problems we will consider in the course

- Transportation problem
- Shortest path
- Time constrained shortest path
- Critical path
- Maximum flow
- Minimum-cost flow
- Multicommodity-flow

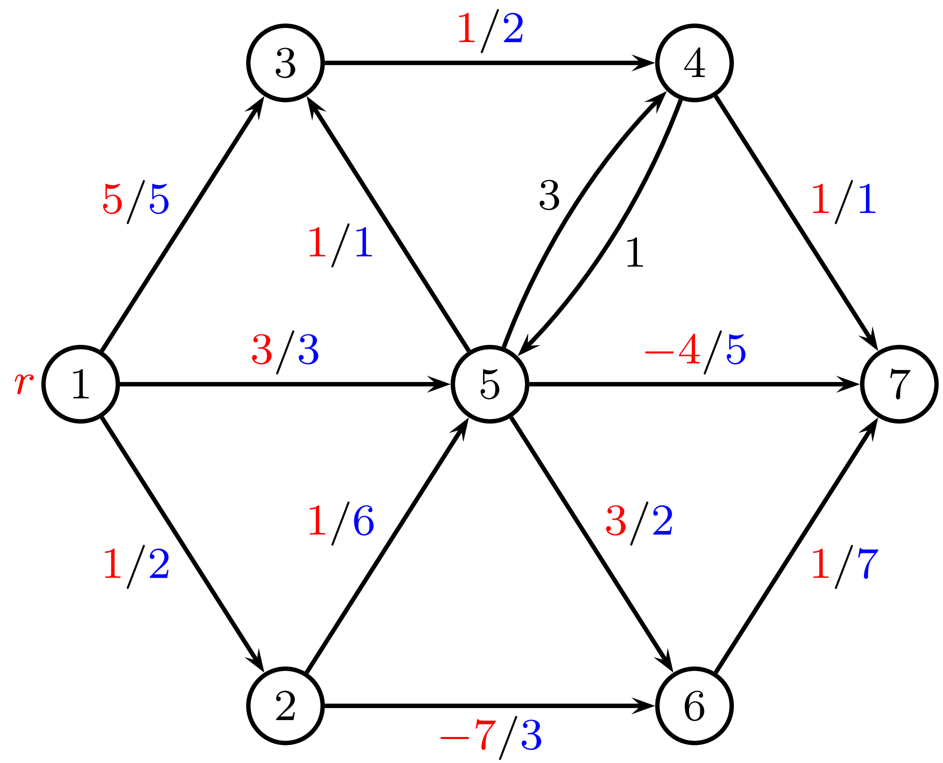
Transportation Problem



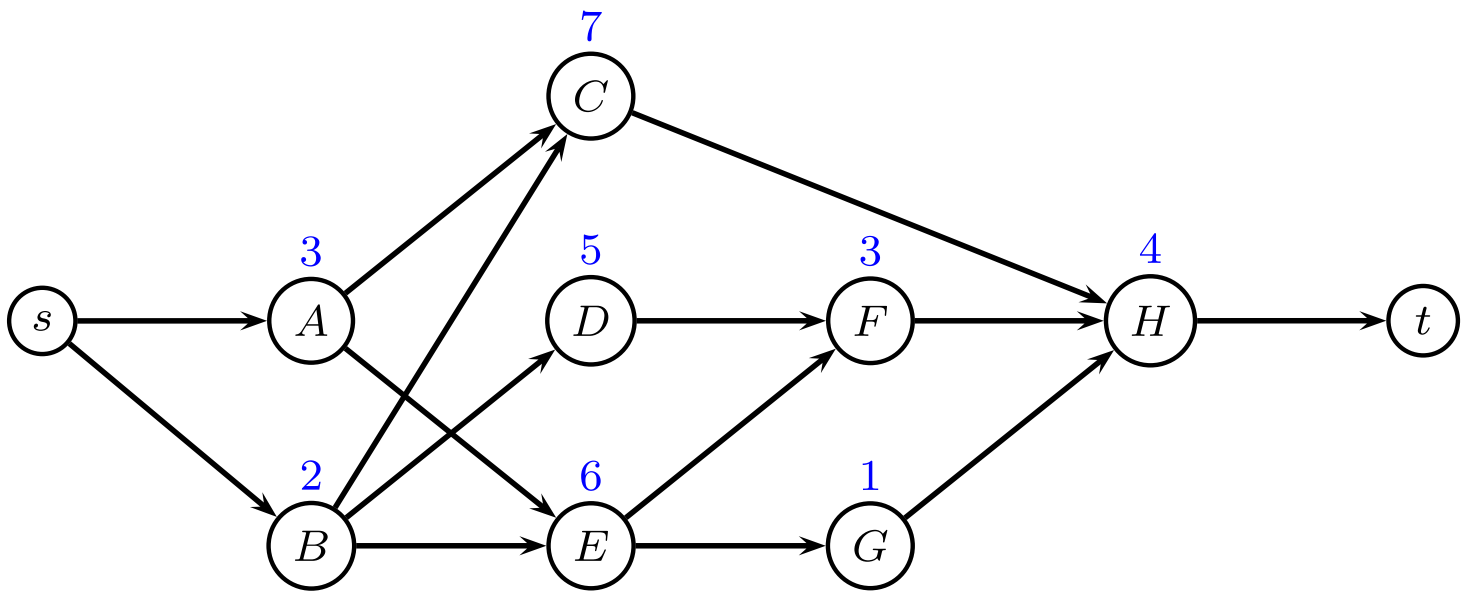
Shortest Path Problem



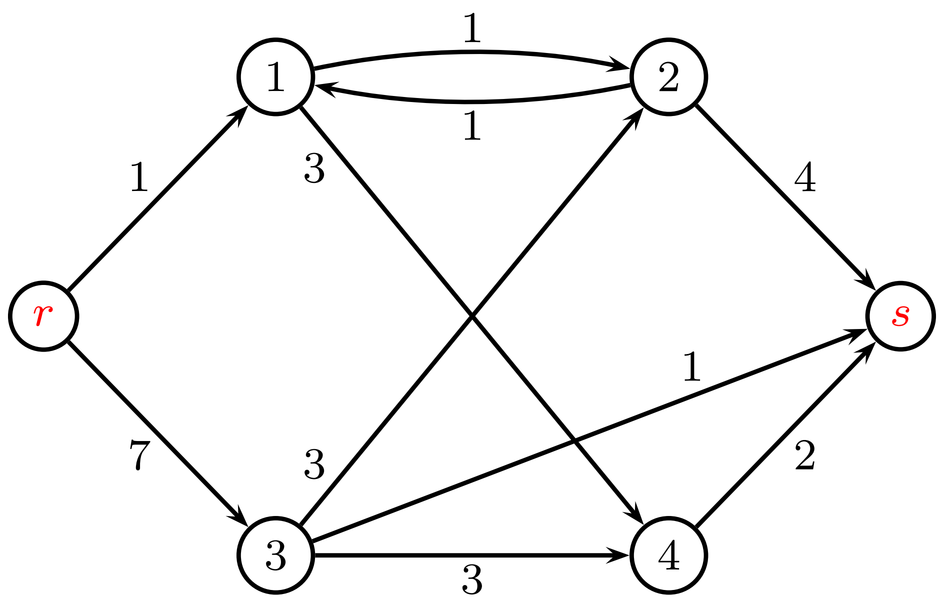
Time-constrained Shortest Path Proble



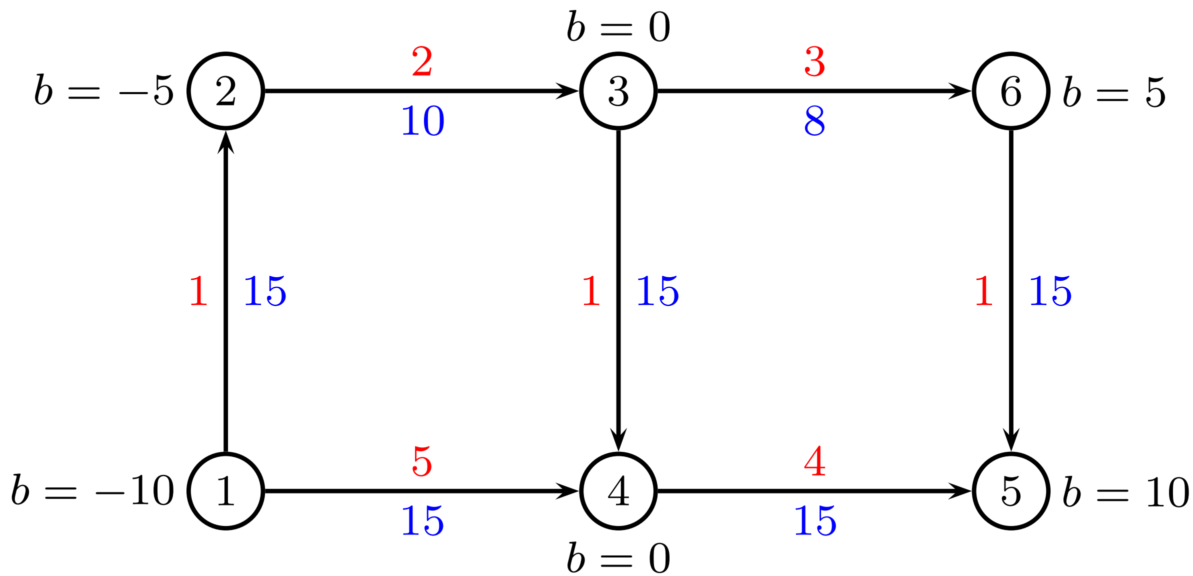
Critical Path Problem



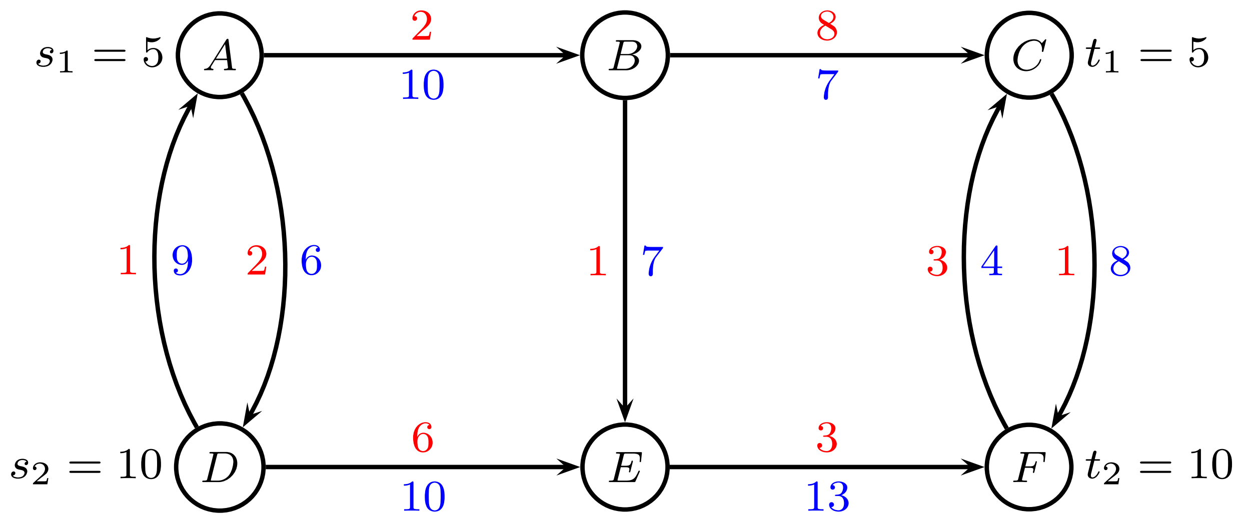
Max Flow Problem



Min Cost Flow



Multicommodity Cost Flow



Critical Path Methods

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Project Management

Consider building a house:

Step A	Prepare site (5 days)	
Step B	Build foundation (8 days)	A
Step C	Frame walls and roof (15 days)	B
Step D	Rough in Plumbing (12 days)	A
Step E	Rough in Electrical (10 days)	A, B
Step F	HVAC Venting (8 days)	A, B, C
Step G	Drywall (11 days)	B, C
Step H	Finish Electrical (5 days)	E, F, G
Step I	Finish Plumbing (4 days)	D, F, G
Step J	Finish HVAC (2 days)	F, G
Step K	Install Kitchen (8 days)	J
Step L	Install Baths (14 days)	I, J
Step M	Paint (5 days)	K, L
Step N	Landscape (5 days)	A, B, G, J

Project Management

What questions might project managers be interested in?

- How long will the project take?
- What tasks are on the critical path?
- Is the project on schedule?
- When should materials and personnel be in place to begin a task?
- Can I add manpower or tools to reduce the overall project length?
- To which tasks should I add manpower?
- Other?

Definitions

- set J of jobs/tasks
- each job $j \in J$ has a processing time p_j
- directed graph $G = (V, E)$. Nodes are jobs, edges state precedences
- no cycles in G
- WLOG, assume J is topologically sorted

Minimize completion time (makespan) T

Topological sorting

Topological sorting of nodes: All edges (i, j) satisfy $i < j$ (i.e. all edges are pointing forward)

Algorithm:

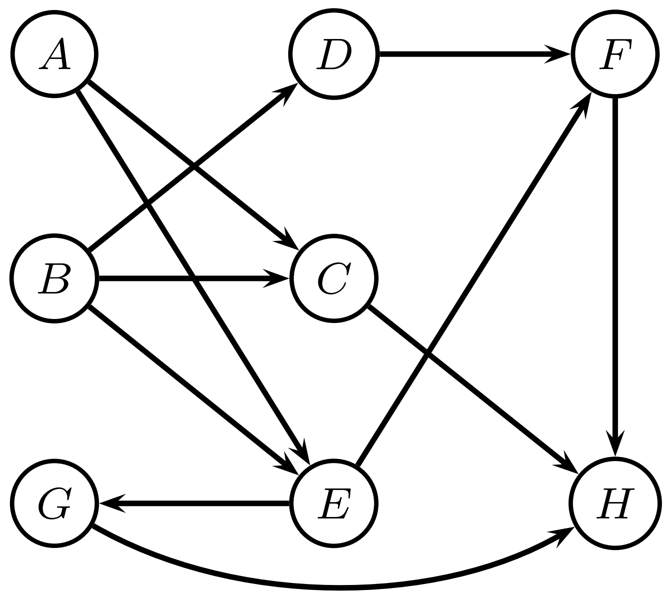
- Given directed acyclic graph $G = (V, E)$
- Repeat
- Since no cycles, there must be a node with no ingoing edges
- Remove this node i and corresponding edges
- Add i to ordering of nodes

Can be done in $O(V + E)$

Small example

j	A	B	C	D	E	F	G	H
p_j	3	2	7	5	6	3	1	4

Precedences:

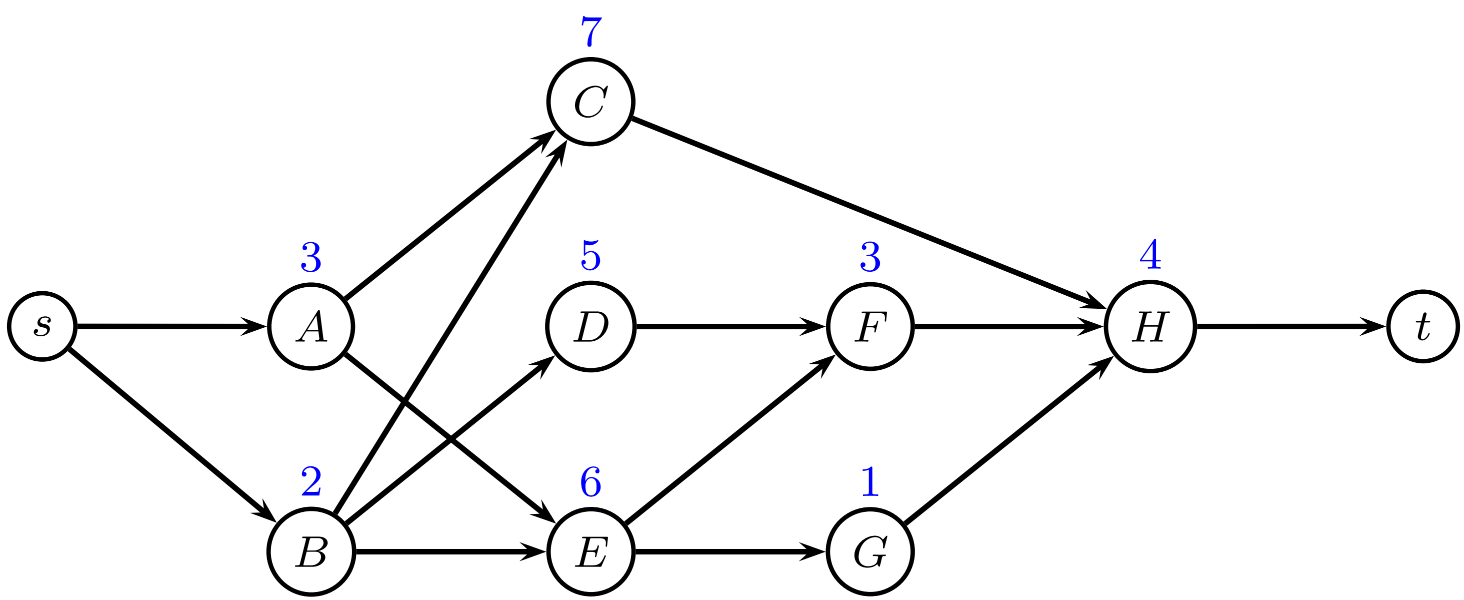


Find makespan, critical path

Small example

j	A	B	C	D	E	F	G	H
p_j	3	2	7	5	6	3	1	4

Topological ordering of precedences:



Find makespan, critical path

Critical Path method

Forward procedure:
Starting at time zero, calculate the earliest time U_j each job j can be started. The completion time of the last job is the makespan T

Backward procedure:
Starting at time equal to the makespan, calculate the latest time V_j each job j can be started so that this makespan is realized

Forward procedure

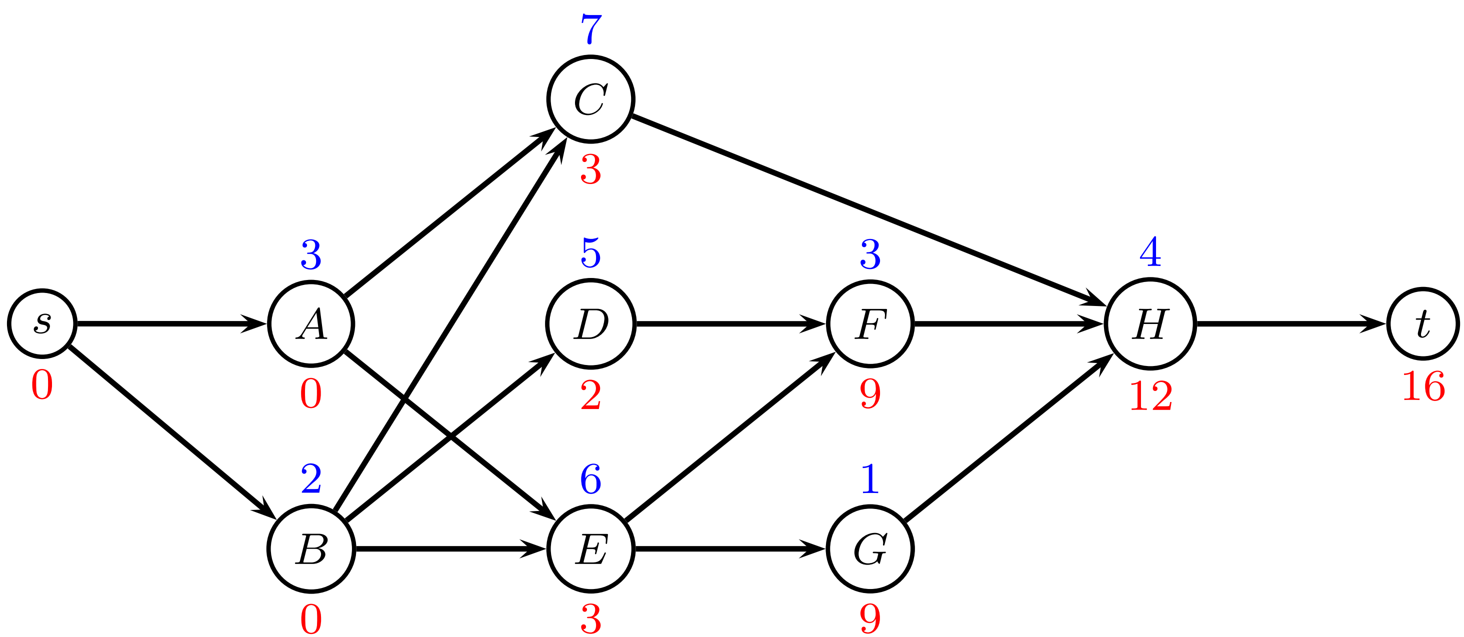
Dynamic programming method:

$U_s = 0$
for $j = 1$ to n
$\quad U_j = \max_{(i,j)\in E}\{U_i + p_i\}$
makespan: $T = U_t$

U_j earliest starting time job j

Forward algorithm

j	A	B	C	D	E	F	G	H
p_j	3	2	7	5	6	3	1	4
U_j	0	0	3	2	3	9	9	12



Makespan: $T = 16$

Backward procedure

Dynamic programming method:

1 $V_t = T$

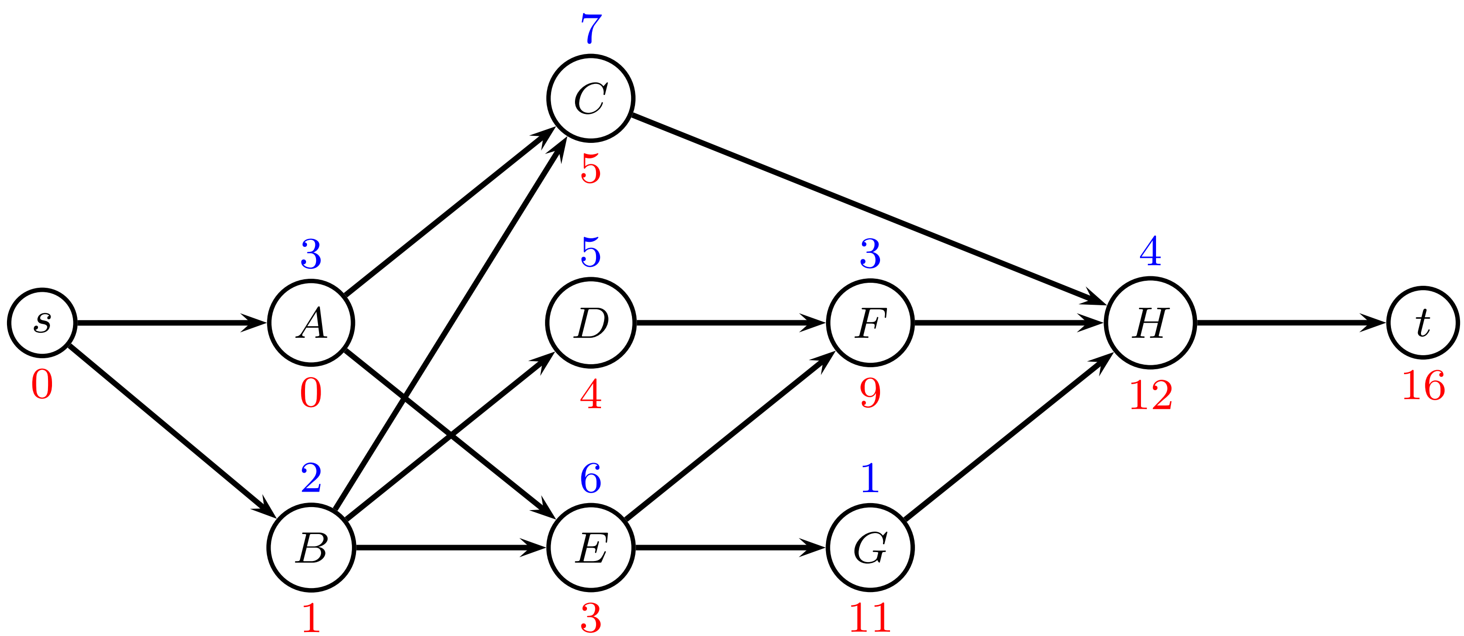
2 for $i = n$ to 1

3 $\quad V_i = \min_{(i,j)\in E}\{V_j - p_i\}$

V_j latest starting time job j

Backward algorithm

j	A	B	C	D	E	F	G	H
p_j	3	2	7	5	6	3	1	4
U_j	0	0	3	2	3	9	9	12
V_j	0	1	5	4	3	9	11	12

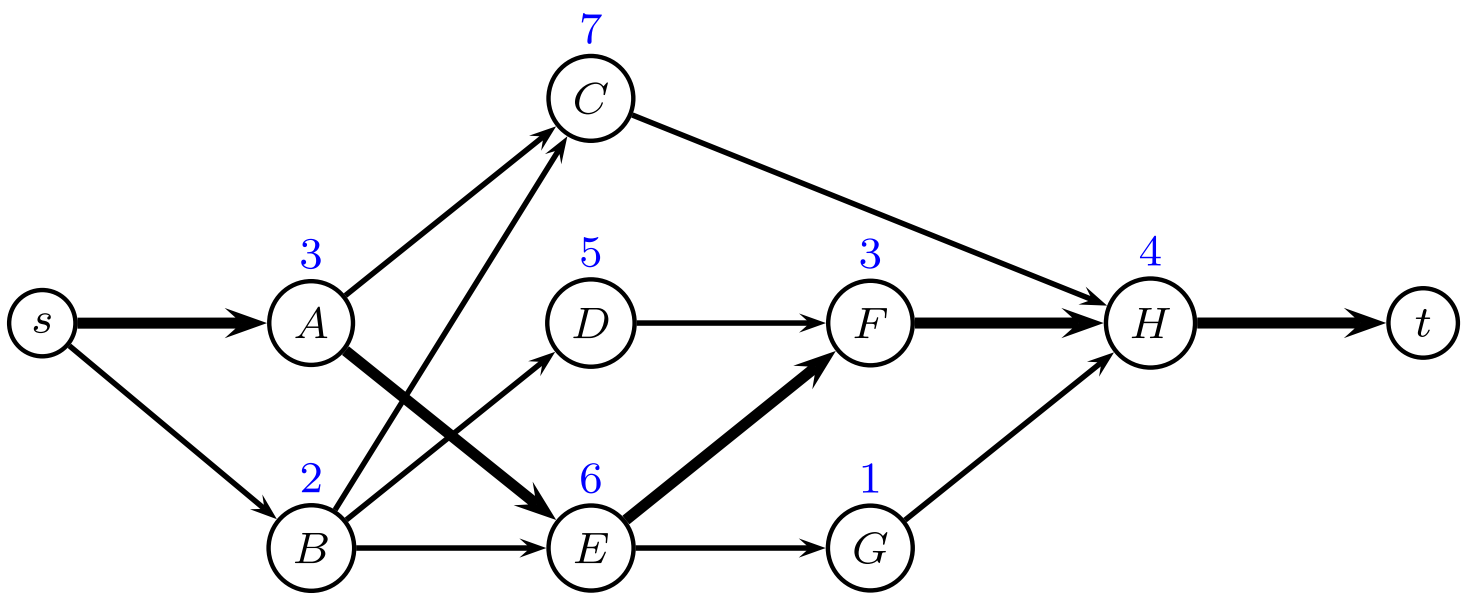


Critical Path

- The forward procedure gives the earliest possible starting time for each job
- The backwards procedures gives the latest possible starting time for each job
- If these are equal the job is a critical job.
- If these are different the job is a slack job, and the difference is the float.
- A critical path is a chain of jobs starting at time 0 and ending at T.

Critical Path

j	A	B	C	D	E	F	G	H
p_j	3	2	7	5	6	3	1	4
U_j	**0**	0	3	2	**3**	**9**	9	**12**
V_j	**0**	1	5	4	**3**	**9**	11	**12**



Nodes with no slack define critical path

LP formulation

Variable x_j is start time of job j

LP-model:

$$\begin{array}{lll} \min & T = x_t - x_s & \\ \text{s.t.} & x_j \geq x_i + p_i & (i,j) \in E \\ & x_j \geq 0 & j \in J \end{array}$$

LP formulation

$$
\begin{aligned}
\min \quad & T = x_t - x_s \\
\text{s.t.} \quad & x_A \geq x_s + 0 \\
& x_B \geq x_s + 0 \\
& x_C \geq x_A + 3 \\
& x_C \geq x_B + 2 \\
& x_D \geq x_B + 2 \\
& x_E \geq x_A + 3 \\
& x_E \geq x_B + 2 \\
& x_F \geq x_D + 5 \\
& x_F \geq x_E + 6 \\
& x_G \geq x_E + 6 \\
& x_H \geq x_C + 7 \\
& x_H \geq x_F + 3 \\
& x_H \geq x_G + 1 \\
& x_t \geq x_H + 4 \\
& x_i \geq 0 \qquad\qquad i \in J
\end{aligned}
$$

LP formulation

```
minimize
    xt - xs
subject to
sA:   xA - xs >= 0
sB:   xB - xs >= 0
AC:   xC - xA >= 3
BC:   xC - xB >= 2
BD:   xD - xB >= 2
AE:   xE - xA >= 3
BE:   xE - xB >= 2
DF:   xF - xD >= 5
EF:   xF - xE >= 6
EG:   xG - xE >= 6
CH:   xH - xC >= 7
FH:   xH - xF >= 3
GH:   xH - xG >= 1
Ht:   xt - xH >= 4
end
```

Running CPLEX

- If you use Windows, install "Thinlinc" so you can connect to another computer. You can download it from `http://gbar.dtu.dk/faq/43-thinlinc`
- Login to GBAR at DTU using Thinlinc. Username is your study number "s123456", while host is "thinlinc.gbar.dtu.dk"
- If you use Linux, you can connect to DTU using `ssh username@thinlinc.gbar.dtu.dk`
- Use an editor, e.g. "gedit" to type in the model
- Save model to a file, e.g. "ex1.lp"
- `cplex` (to start CPLEX)

Output from CPLEX

```
CPLEX> read model1
Problem 'model1.lp' read.
Read time =    0.01 sec.
CPLEX> opt
Tried aggregator 1 time.
LP Presolve eliminated 3 rows and 1 columns.
Aggregator did 5 substitutions.
Reduced LP has 6 rows, 4 columns, and 12 nonzeros.
Presolve time =    0.00 sec.
Initializing dual steep norms . . .

Iteration log . . .
Iteration:     1   Dual infeasibility =            0.000000
Iteration:     2   Dual objective     =           14.000000

Dual simplex - Optimal:  Objective =  1.6000000000e+01
Solution time =    0.00 sec.  Iterations = 3 (1)
```

CPLEX solution

```
CPLEX> dis sol var -
Variable Name              Solution Value
xt                              16.000000
xC                               5.000000
xD                               4.000000
xE                               3.000000
xF                               9.000000
xG                              11.000000
xH                              12.000000
All other variables in the range 1-10 are 0.
CPLEX> dis sol dua -
Constraint Name                Dual Price
sA                               1.000000
AE                               1.000000
EF                               1.000000
FH                               1.000000
Ht                               1.000000
All other dual prices in the range 1-14 are 0.
```

If dual variable is positive, edge is on critical path

Hints on using CPLEX

- Use an editor to write the CPLEX input file, e.g. gedit
- Remember to move to the same directory as the file before starting CPLEX, this can be done with the command `cd directoryname`
- All variables must be on the left side of the inequality
- All constants must be on the right side of the inequality
- No constant is allowed in the objective
- No paranteses are allowed, simplify the equation by hand
- No multiplication sign is needed, just write `2 x1`
- All the output from the screen is also written to a file `cplex.log` which is found in the same directory. You can copy-paste otput from this file to your assignments.

Project Management

- Can normal task times be reduced?
- Is there an increase in direct costs? Additional manpower, Additional machines, Overtime, etc
- Can there be a reduction in indirect costs? Less overhead costs, Less daily rental charges, Bonus for early completion, Avoid penalties for running late, Avoid cost of late startup

CPM addresses these cost trade-offs.

Crashing project

j	A	B	C	D	E	F	G	H
p_j	3	2	7	5	6	3	1	4
m_j	2	1	4	4	6	2	1	3
c_j	4	3	1	2	7	5	3	6

- p_j normal process time
- m_j minimal process time
- c_j cost of decreasing time by one unit

LP formulation, model 1

Variable x_i is start time of taks i
Variable y_i is duration of task i

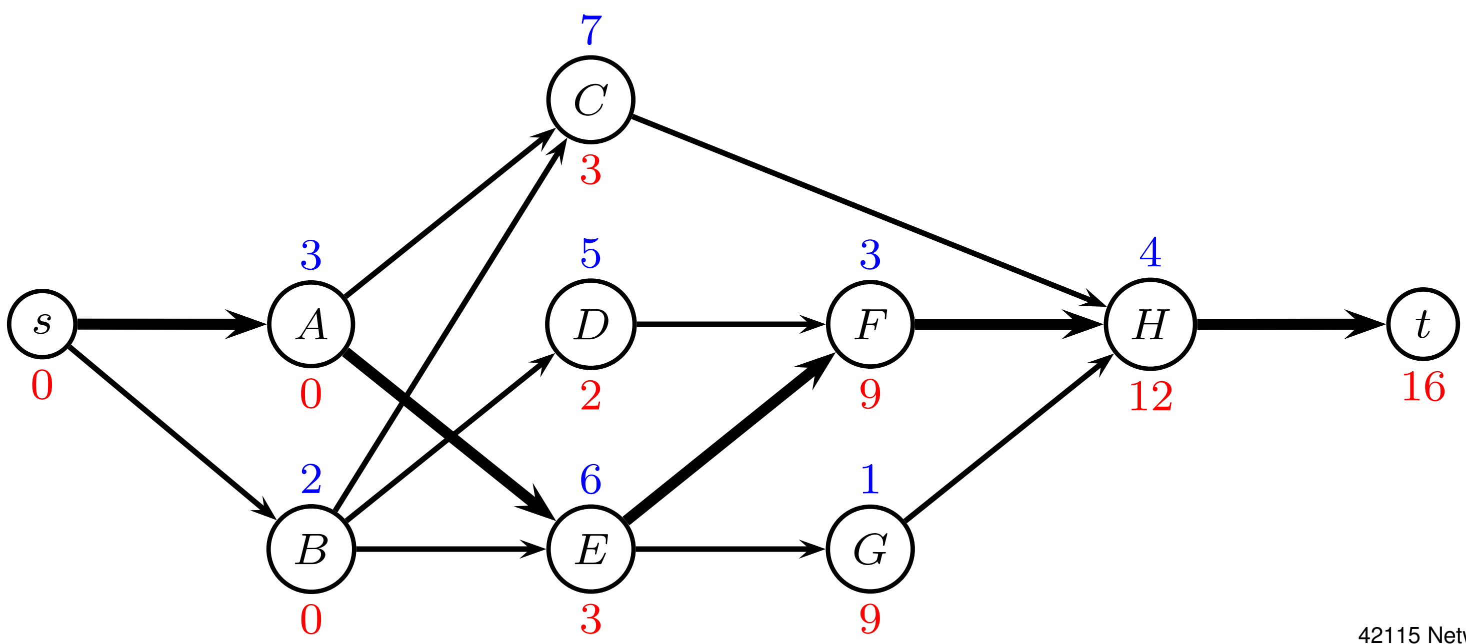
Model 1: cheapest way of finishing within time T

$$
\begin{array}{lll}
\min & \sum_{j \in J} c_j(p_j - y_j) & \\
& x_j \geq x_i + y_i & (i,j) \in E \\
& x_t - x_s \leq T & \\
& m_j \leq y_j \leq p_j & j \in J \\
& x_j \geq 0 & j \in J
\end{array}
$$

LP solution, T=16

j	A	B	C	D	E	F	G	H
p_j	3	2	7	5	6	3	1	4
m_j	2	1	4	4	6	2	1	3
c_j	4	3	1	2	7	5	3	6
x_j	0	0	5	4	3	9	11	12
y_j	3	2	7	5	6	3	1	4

cost: 0



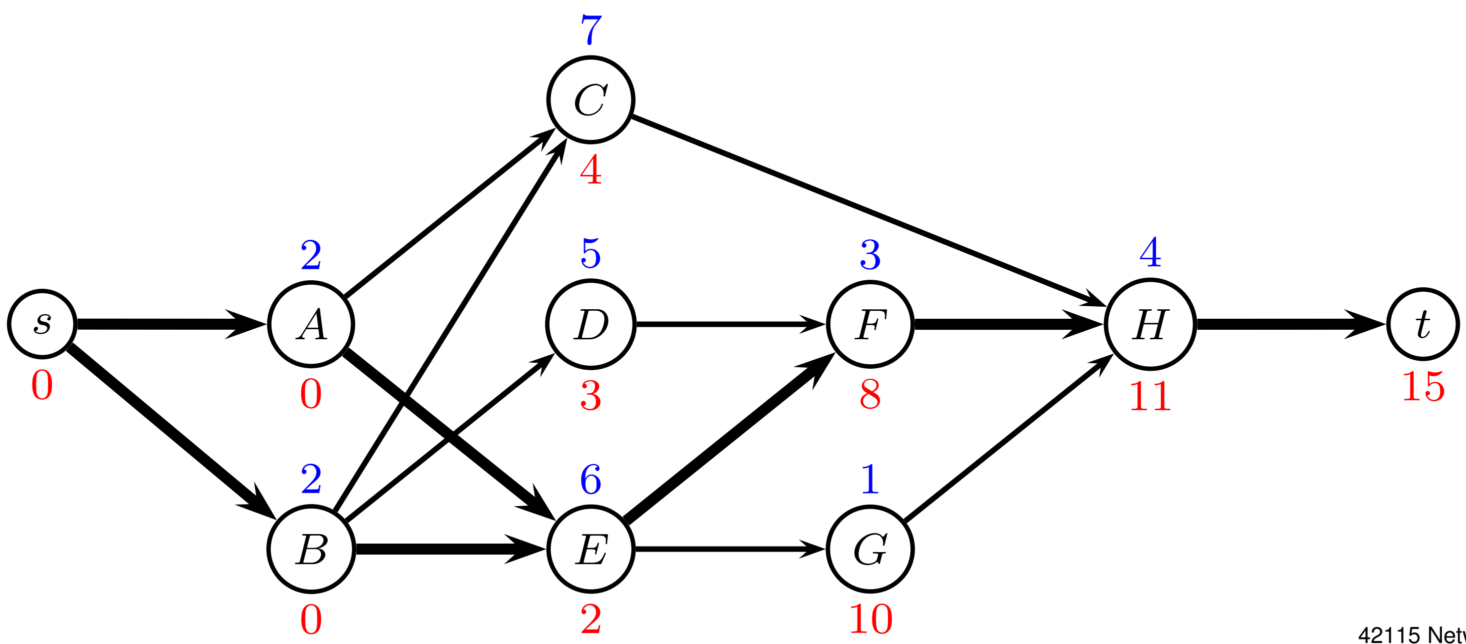
LP solution, T=16

Variable Name	Solution Value
yA	3.000000
yB	2.000000
yC	7.000000
yD	5.000000
yE	6.000000
yF	3.000000
yG	1.000000
yH	4.000000
xC	5.000000
xD	4.000000
xE	3.000000
xF	9.000000
xG	11.000000
xH	12.000000
xt	16.000000

LP solution, T=15

j	A	B	C	D	E	F	G	H
p_j	3	2	7	5	6	3	1	4
m_j	2	1	4	4	6	2	1	3
c_j	4	3	1	2	7	5	3	6
x_j	0	0	4	3	2	8	10	11
y_j	2	2	7	5	6	3	1	4

cost: 4



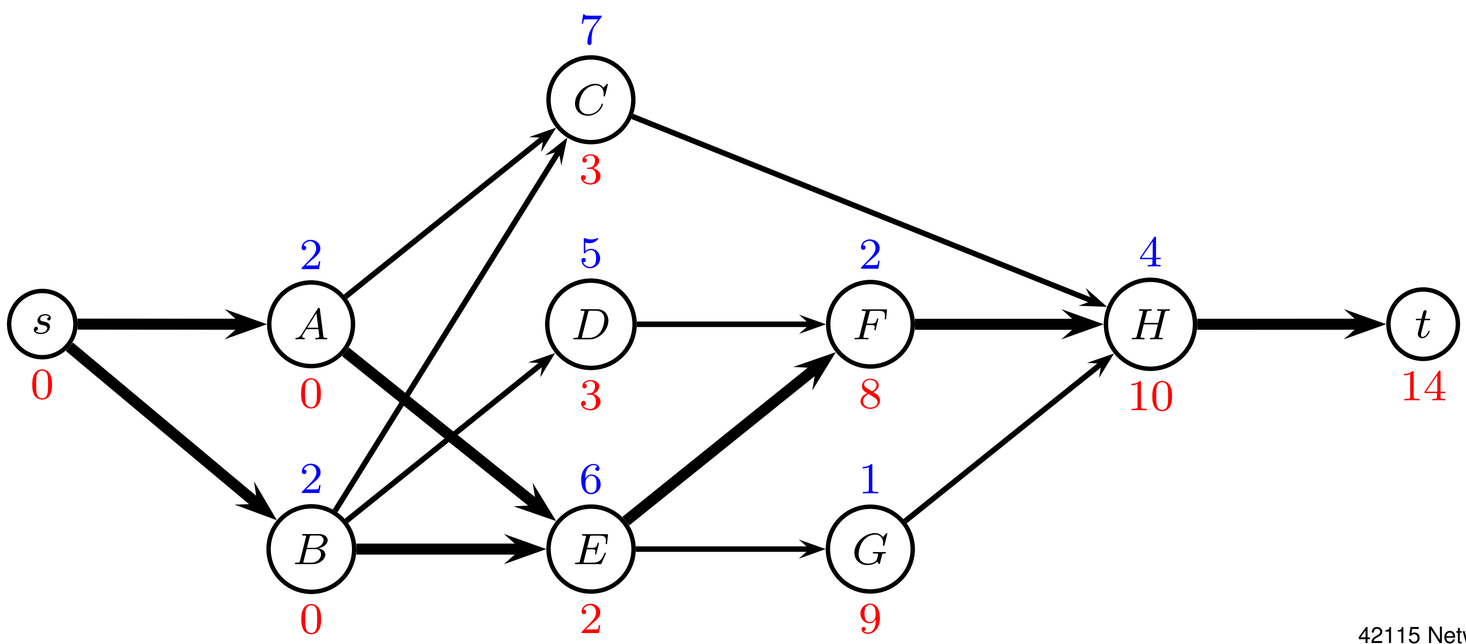
LP solution, T=15

yA	2.000000
yB	2.000000
yC	7.000000
yD	5.000000
yE	6.000000
yF	3.000000
yG	1.000000
yH	4.000000
xC	4.000000
xD	3.000000
xE	2.000000
xF	8.000000
xG	10.000000
xH	11.000000
xt	15.000000

LP solution, T=14

j	A	B	C	D	E	F	G	H
p_j	3	2	7	5	6	3	1	4
m_j	2	1	4	4	6	2	1	3
c_j	4	3	1	2	7	5	3	6
x_j	0	0	3	3	2	8	9	10
y_j	2	2	7	5	6	2	1	4

cost: 9



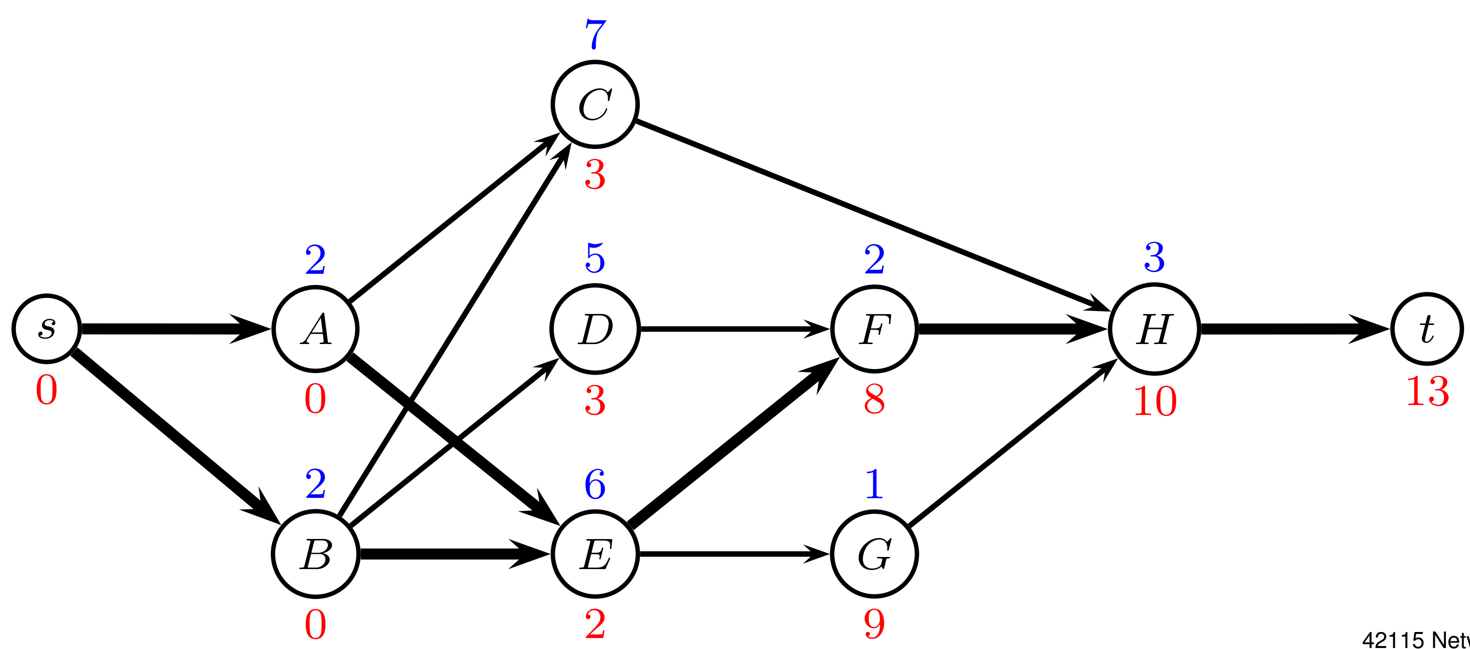
LP solution, T=14

yA	2.000000
yB	2.000000
yC	7.000000
yD	5.000000
yE	6.000000
yF	2.000000
yG	1.000000
yH	4.000000
xC	3.000000
xD	3.000000
xE	2.000000
xF	8.000000
xG	9.000000
xH	10.000000
xt	14.000000

LP solution, T=13

j	A	B	C	D	E	F	G	H
p_j	3	2	7	5	6	3	1	4
m_j	2	1	4	4	6	2	1	3
c_j	4	3	1	2	7	5	3	6
x_j	0	0	3	3	2	8	9	10
y_j	2	2	7	5	6	2	1	3

cost: 15



LP solution, T=13

cost: 15

yA	2.000000
yB	2.000000
yC	7.000000
yD	5.000000
yE	6.000000
yF	2.000000
yG	1.000000
yH	3.000000
xC	3.000000
xD	3.000000
xE	2.000000
xF	8.000000
xG	9.000000
xH	10.000000
xt	13.000000

LP solution, T=12

Impossible

All tasks on critical path have minimum processing time

LP formulation, model 2

Variable x_i is start time of taks i
Variable y_i is duration of task i

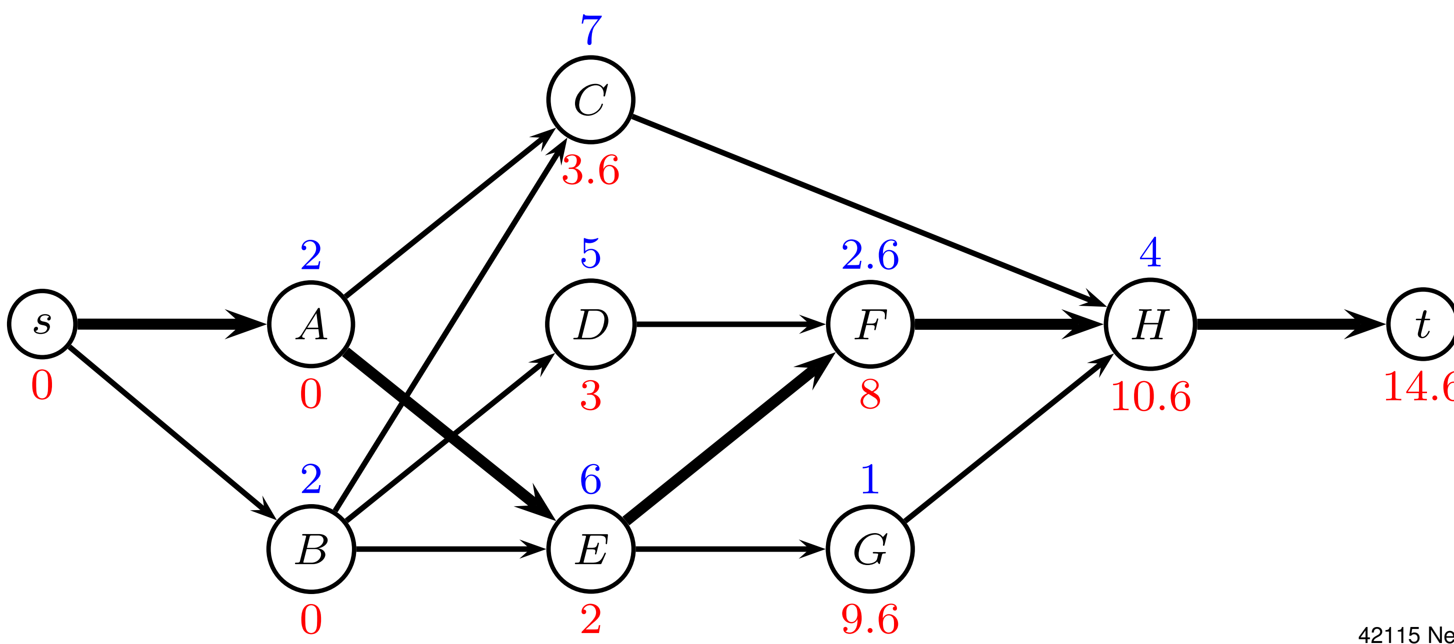
Model 2: shortest completion time, given budget B

$$\begin{array}{lll}
\min & x_t - x_s & \\
& x_j \geq x_i + y_i & (i,j) \in E \\
& \sum_{j \in J} c_j (p_j - y_j) \leq B & \\
& m_j \leq y_j \leq p_j & j \in J \\
& x_j \geq 0 & j \in J
\end{array}$$

LP solution, B=6

j	A	B	C	D	E	F	G	H
p_j	3	2	7	5	6	3	1	4
m_j	2	1	4	4	6	2	1	3
c_j	4	3	1	2	7	5	3	6
x_j	0	0	3.6	3	2	8	9.6	10.6
y_j	2	2	7	5	6	2.6	1	4

T: 14.6



LP solution, B=6

T: 14.6

xt	14.600000
xC	3.600000
yA	2.000000
yB	2.000000
xD	3.000000
xE	2.000000
xF	8.000000
yD	5.000000
yE	6.000000
xG	9.600000
xH	10.600000
yC	7.000000
yF	2.600000
yG	1.000000
yH	4.000000

Resource constraints

Renewable resources are available on a period-by-period basis, i.e. the available amount is renewed from period to period.
Typical examples are manpower, machines, tools, equipment and space.
Let R'_i be the amount of resource i
Let R_{ij} be the amount of resource i used by job j

Resource constraints

- No LP formulation
- Can develop an IP
- Hard to solve

Let job t be dummy job (sink) and

$$x_{j\tau} = \left\{ \begin{array}{ll} 1 & \text{if job } j \text{ is completed at time } \tau \\ 0 & \text{otherwise} \end{array} \right\}$$

Resource constraints

Let H be an upper bound on the makespan, e.g.

$$H = \sum_{j=1}^{n} p_j$$

Completion time of job j is

$$\sum_{\tau=1}^{H} \tau x_{j\tau}$$

Makespan is

$$T = \sum_{\tau=1}^{H} \tau x_{t,\tau}$$

Project Management

$$
\begin{array}{lll}
\min & \sum_{\tau=1}^{H} \tau x_{t,\tau} & \\
\text{s.t.} & \sum_{\tau=1}^{H} \tau x_{j\tau} + p_k - \sum_{\tau=1}^{H} \tau x_{k\tau} \leq 0 & (j,k) \in E \\
& \sum_{\tau=1}^{H} \left(R_{ij} \sum_{u=\tau}^{\tau+p_j-1} x_{ju} \right) \leq R_i' & \text{each resource used} \\
& \sum_{\tau=1}^{H} x_{j\tau} = 1 & j \in J
\end{array}
$$

- minimize completion time of last job
- precedence constraints
- resource constraint constraints
- job must finish at exactly one time